Supplementary Material for "Phase-tuned delayed feedback for simultaneous linewidth and relative intensity noise suppression in semiconductor lasers"

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S1. External-cavity mode and linearized matrices

For $E(t) = Ae^{\mathrm{i}\Omega t}$ and $N = N_s$, define $\theta = \phi + \Omega\tau$. The ECM equations are

$$\theta - \phi + \kappa\tau(\alpha\cos\theta + \sin\theta) = 0, \tag{1}$$

$$N_s = -\kappa\cos\theta, \tag{2}$$

$$A^2 = \frac{P - N_s}{1 + 2N_s}, \tag{3}$$

$$\Omega = -\kappa(\alpha\cos\theta + \sin\theta). \tag{4}$$

Let $c_r = \kappa\cos\theta$ and $c_i = -\kappa\sin\theta$. In the rotating frame, the matrices in $\dot{x} = A_0 x + A_1 x(t-\tau) + B\eta$ are

$$A_0 = \begin{pmatrix} -c_r & c_i & A \\ -c_i & -c_r & \alpha A \\ -2\epsilon(1+2N_s)A & 0 & -\epsilon(1+2A^2) \end{pmatrix}, \qquad A_1 = \begin{pmatrix} c_r & -c_i & 0 \\ c_i & c_r & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{5}$$

The characteristic matrix is $\mathbf{\Delta}(\lambda) = \lambda I - A_0 - A_1 e^{-\lambda\tau}$.

S2. Weak-feedback sensitivity

At $\kappa = 0$, $A = \sqrt{P}$, $N_s = 0$, and the first derivative of the ECM intensity is

$$\left.\frac{dA^2}{d\kappa}\right|_0 = (1 + 2P)\cos\phi. \tag{6}$$

Differentiation of the exact phase-diffusion ratio gives

$$g_\varphi(\phi) = -\frac{1+2P}{P}\cos\phi - 2\tau(\cos\phi - \alpha\sin\phi). \tag{7}$$

For RIN, write $H_0(\mathrm{i}\omega) = (\mathrm{i}\omega I - A_0^{(0)})^{-1}B$. With

$$K_\phi(\omega) = A_{0,\kappa} + e^{-\mathrm{i}\omega\tau}A_{1,\kappa}, \tag{8}$$

we use

$$\partial_\kappa H_0 = H_0 K_\phi H_0. \tag{9}$$

The output row $C_{\mathrm{RIN}} = (2/A, 0, 0)$ is also differentiated. If $G_0 = C_0 H_0$ and $G_\kappa = C_\kappa H_0 + C_0 \partial_\kappa H_0$, then

$$g_{\mathrm{RIN}}(\phi) = \frac{2\,\mathrm{Re}\int_0^\infty \mathrm{tr}[G_\kappa(\omega) G_0(\omega)^*]\,d\omega}{\int_0^\infty \mathrm{tr}[G_0(\omega) G_0(\omega)^*]\,d\omega}. \tag{10}$$

For the baseline parameters, numerical quadrature gives

$$g_\varphi = -43.000000\cos\phi + 160.000000\sin\phi, \tag{11}$$

$$g_{\mathrm{RIN}} = -106.862441\cos\phi - 344.419746\sin\phi. \tag{12}$$

The maximum residual from a first-harmonic fit is 2.8×10^{-12}. Table S1 audits the formulas against finite differences at $\kappa = 10^{-5}$.

Table 1: First-order sensitivity audit.

ϕ	g_φ formula	finite difference	g_{RIN} formula	finite difference
-0.25	-81.248	-81.304	-18.330	-18.083
0	-43.000	-43.115	-106.862	-106.472
0.20	-10.356	-10.491	-173.158	-172.577
0.50	38.972	38.855	-258.904	-257.981

S3. Phase diffusion and residue normalization

For a simple characteristic root λ_j, the resolvent has the local expansion

$$\boldsymbol{\Delta}(s)^{-1} = \frac{v_j u_j^*}{u_j^* \boldsymbol{\Delta}'(\lambda_j) v_j} \frac{1}{s-\lambda_j} + O(1). \tag{13}$$

At the neutral root, projection of white input noise gives

$$D_\varphi = \frac{1}{2} \frac{u_0^* Q u_0}{|u_0^* \boldsymbol{\Delta}'(0) v_0|^2}. \tag{14}$$

At a nonzero conjugate pair $\lambda_\pm$, integration of the retained transfer function gives

$$V_y^{(\pm)} = -\sum_{j,k\in\{+,-\}} \frac{\mathrm{tr}(C R_j B B^* R_k^* C^*)}{\lambda_j + \lambda_k^*}. \tag{15}$$

The off-diagonal terms are retained. The pair reproduces 98.78% of the trade-off RIN in the relaxation band and 94.49% of the simultaneous-suppression RIN.

S4. Characteristic-root audit

Roots were first seeded with an upwind transport-chain representation of the delay and then refined directly as zeros of $\det \boldsymbol{\Delta}(\lambda)$. The argument principle was evaluated on the rectangle

$$-0.05 < \mathrm{Re}\,\lambda < 0.02, \qquad |\mathrm{Im}\,\lambda| < 0.5. \tag{16}$$

For the solitary, trade-off, and simultaneous-suppression states, the winding number is three, corresponding to the neutral root and one conjugate relaxation pair. The minimum determinant magnitude on the contour is respectively 2.14×10^{-4}, 2.92×10^{-4}, and 2.44×10^{-4}. Table S2 shows convergence of the leading transport-chain root.

Table 2: Transport-chain convergence. Values are the root in the lower half-plane.

State	chain cells	$\mathrm{Re}\,\lambda$	$\mathrm{Im}\,\lambda$
Trade-off	30	−0.003908	−0.104139
	60	−0.003923	−0.104237
	120	−0.003932	−0.104286
Direct refinement	–	−0.003944	−0.104336
Simultaneous suppression	30	−0.015469	−0.093979
	60	−0.015455	−0.093765
	120	−0.015443	−0.093658
Direct refinement	–	−0.015428	−0.093550

S5. Nonlinear stochastic delay simulations

The full nonlinear equations were integrated in the ECM rotating frame using Euler–Maruyama. The delay was stored in a circular buffer with $\Delta t = 0.02$, the record spacing was 0.1, and 64 independent realizations were simulated for $T = 3000$ after a burn-in of 400. The complex field noise amplitude was $\sigma = 5 \times 10^{-4}$ in each quadrature. RIN and carrier variance were estimated after burn-in. Phase diffusion was estimated from time- and ensemble-averaged unwrapped-phase increments over lags from 150 to 780 normalized time units. Uncertainty bars in the main text are two standard errors across realizations.

Table 3: Direct nonlinear simulation and linear-resolvent ratios.

	linewidth		RIN		carrier	
State	simulation	linear	simulation	linear	simulation	linear
Trade-off	0.509	0.471	1.444	1.565	1.323	1.438
Simultaneous suppression	0.824	0.750	0.478	0.480	0.514	0.518

The stationary RIN and carrier results are insensitive to halving the time step within sampling uncertainty. The phase estimate is the slowest-converging quantity because it depends on long increments and contains bounded relaxation-mode contamination at finite lag.

S6. Noise-source, bandwidth, and fixed-output tests

The field-noise matrix is

$$B_E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{17}$$

A carrier-current source is represented by $B_N = (0, 0, 1)^T$. For the balanced mixed model, its scale is chosen so that field and carrier sources make equal contributions to the solitary integrated RIN; this gives a carrier-column scale of $\sqrt{0.0102245}$. Table S4 reports RIN ratios.

For normalized upper detection bandwidths 0.15, 0.25, 0.5, 3, and 200, the trade-off ratios are 1.622, 1.590, 1.576, 1.566, and 1.565, whereas the simultaneous-suppression ratios are 0.428, 0.458, 0.470, 0.478, and 0.480.

Table 4: Integrated RIN ratios for alternative source models.

Source model	Trade-off	Simultaneous suppression
Field Langevin source	1.565	0.480
Balanced field + carrier source	1.744	0.490

At fixed output power $A = 1$, the pump is adjusted to $P = 1 + 3N_s$. The results are shown in Fig. S1.

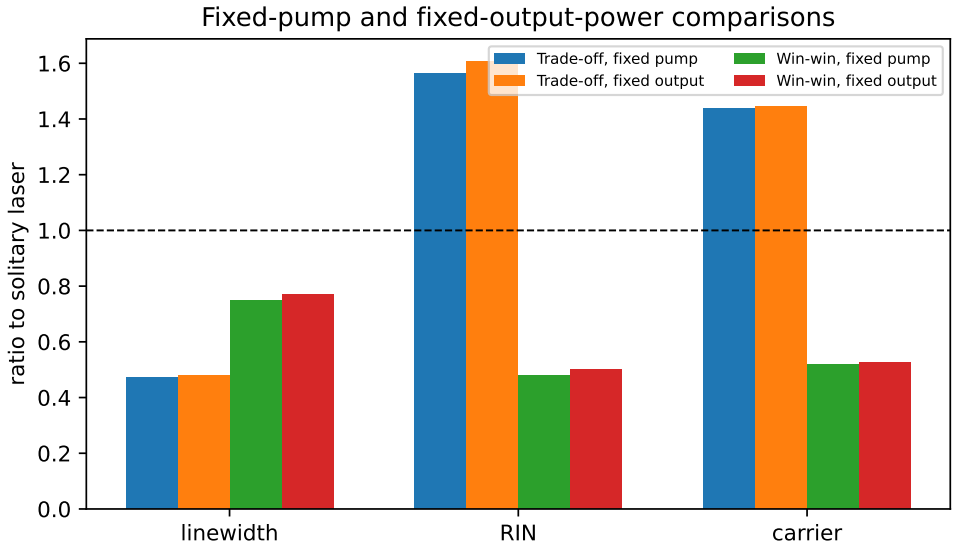


Figure 1: Comparison at fixed pump and fixed output power. The linewidth–RIN classification is unchanged.

S7. Extended robustness

A coarse scan used $0 < \kappa \leq 0.009$ and 37 phases over $[-\pi, \pi]$. Gain compression was introduced through $G = N/(1 + \eta|E|^2)$ in the field equation and the corresponding saturated stimulated-emission term in the carrier equation. Table S5 lists the best stable simultaneous-suppression point for each configuration.

Table 5: Extended robustness scan.

Configuration	linewidth ratio	RIN ratio	stability margin	stable win–win fraction
$P = 0.7$	0.681	0.507	0.0105	0.162
$P = 1.0$	0.684	0.524	0.0137	0.153
$P = 1.5$	0.686	0.567	0.0185	0.141
$\epsilon = 0.003$	0.684	0.467	0.00834	0.160
$\epsilon = 0.008$	0.684	0.606	0.0211	0.135
$\eta = 0.05$	0.681	0.550	0.0128	0.144

Configuration	linewidth ratio	RIN ratio	stability margin	stable win–win fraction
$\eta = 0.10$	0.679	0.579	0.0118	0.142

Additional searches over $\alpha \in \{2, 4, 6\}$ and $\tau \in \{10, 20, 40\}$ found at least one stable simultaneous-suppression state in every combination. The selected phase shifts with α, as expected from amplitude–phase coupling.

S8. Device-scale illustration

A photon lifetime $\tau_p = 2.5$ ps is used only to demonstrate the dimensional conversion. Then $\epsilon = 0.005$ corresponds to a carrier lifetime of 500 ps, $\tau = 20$ corresponds to an external round trip of 50 ps, and normalized angular frequency maps to $f = \omega/(2\pi\tau_p)$. The white-noise strength is calibrated so that the solitary Lorentzian linewidth is 2 MHz. The resulting values are:

Table 6: Illustrative dimensional values.

State	linewidth (MHz)	RIN-peak frequency (GHz)	RIN peak (dBc/Hz)
Solitary	2.000	6.366	−130.76
Trade-off	0.941	6.641	−126.07
Simultaneous suppression	1.500	6.048	−137.20

These numbers are not a fit to a particular device. They show that the normalized ratios correspond to realistic semiconductor-laser time scales and to directly measurable linewidth and RIN quantities.

S9. Reproducibility files

The archive contains the ECM solver, exact delay-resolvent quadrature, phase-diffusion calculation, characteristic-root refinement, argument-principle audit, weak-feedback sensitivity, nonlinear stochastic delay simulations, noise-model and bandwidth tests, gain-compression scans, dimensional conversion, numerical tables, and all figure scripts. CSV files report every number used in the main text and this supplement.